Republic of the Philippines
Department of Environment and Natural Resources
MINES AND GEOSCIENCES BUREAU
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FLOOD THREAT ADVISORY

To: Brgy. Chairperson
Brgy. Santo Niño
City of Marikina
Metropolitan Manila

Date: October 9, 2024

Dear Sir/Madam:

Please be advised that the Geohazards Mapping and Assessment Team (GMAT) of the Mines and Geosciences Bureau (MGB) has conducted **flood assessment** for the updating of 1:10,000 Scale Geohazard Map in your barangay. The following are the results and recommendations following the assessment:

Purok / Sitio / Area of Concern	GPS Coordinates (Luzon 1911)	Flood Susceptibility Rating	Geohazards Assessment
Barangay Hall and vicinity	14° 38' 30.4" N, 121° 05' 43.1" E	High	The barangay hall and its vicinity experience a range of flood heights from less than 0.5m to greater than 1m.
Sheff-Aquilina	14° 38' 35.7" N, 121° 06' 02.5" E	High	The Sheff-Aquilina intersection and its vicinity experiences flooding of heights ranging from less than 0.5 to greater than 1m, with maximum height reported during Ondoy.
Amang Rodriguez Hospital	14° 38' 13.6" N, 121° 05' 48.8" E	High	The vicinity of Amang Rodriguez Hospital experienced maximum flood height of ~1.4m during Ondoy which also inundated the interior of the hospital.
Riverbank	14° 38' 14.8" N, 121° 05' 31.5" E	Very High	Marikina River traverses the western boundary of the barangay and during Onday, the 40-ft statue was totally submerged, while ~0.4m-high floodwaters were reported during Ulysses.

Remarks / Recommendations:

- ✓ It is highly recommended to implement appropriate easements in construction of settlements or other infrastructure along rivers/creeks as specified in the Presidential Decree No. 1067 known as Water Code of the Philippines.
- ✓ Improvement and proper maintenance of existing drainage system in the barangay are recommended to prevent inundation of residential areas, critical facilities, and other assets. Avoid improper waste disposal, particularly along drainage canals as it may cause clogging and obstruction of water flow. Ensure regular clearing of natural and artificial drainage pathways, or prior to an impending tropical cyclone or other rain-bearing weather systems.
- ✓ Construction of appropriate and well-studied engineering measures (e.g., dike, flood wall, etc.) should be implemented to minimize the rate of erosion along rivers/creeks. Ensure that the structure follows engineering standards and has structural integrity. Also, damages in the structural mitigation measures should be reported to concerned agency/office responsible for the installation. The LGU is also encouraged to maintain such measures or reinforce identified damages, with the strong support of the barangay officials and active engagement of the communities and the private sector.

- ✓ Install a warning signage alerting motorists and residents of the danger of crossing bridges susceptible to flash flood during heavy rainfall events.
- ✓ The CDRRMO, together with the BDRRMO, should conduct Information, Education, and Communication (IEC) Campaign on flood hazards to flood-prone communities, particularly the households residing proximal to the river and irrigation canals.
- ✓ Strengthen and improve flood warning system with corresponding warning levels for pre-emptive evacuation purposes. The barangay should have constant communication with nearby and/or upland/upstream barangays during typhoon and monsoon season. This is to gather information from their counterparts in terms of the flood situation in their areas.
- ✓ Adhere to the warnings given by proper authorities prior to the landfall or passing of tropical cyclones and other prevailing weather disturbances. This may include City Disaster Risk Reduction Management Office (CDRRMO), Office of the Civil Defense (OCD) NCR, and other related agencies.

Prepared by:

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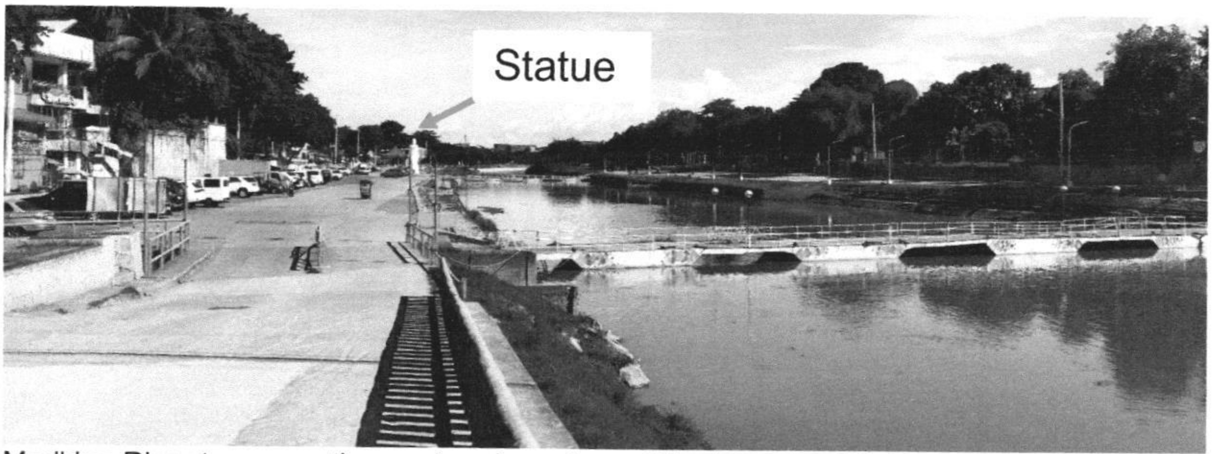
Received by:
Designation:
Office:

BARANGAY SANTO NIÑO

The barangay hall *(left)* and its vicinity experience a range of flood heights from less than 0.5m to greater than 1m. Likewise, the Sheff-Aquilina intersection *(right)* and its vicinity experiences flooding of heights ranging from less than 0.5 to greater than 1m, with maximum height reported during Ondoy (GPS Coordinates: 14° 38' 30.4 and 35.7" N, 121° 05 and 06' 43.1 and 02.5" E).

The vicinity of Amang Rodriguez Hospital experienced maximum flood height of ~1.4m during Ondoy which also inundated the interior of the hospital (GPS Coordinates: 14° 38' 13.6" N, 121° 05' 48.8" E).



Marikina River traverses the western boundary of the barangay and during Onday, the 40-ft statue was totally submerged, while ~0.4m-high floodwaters were reported during Ulysses (GPS Coordinates: 14° 38' 14.8" N, 121° 05' 31.5" E).

Recommendations:

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